

SURVEY

Multimodal Learning Techniques for Time Series Forecasting in Renewable Energy Systems: A Comprehensive Survey

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ABSTRACT Renewable energy systems, such as solar, wind, and hybrid sources, present complex forecasting challenges due to their stochastic and weather-dependent nature. The growing availability of heterogeneous and complementary data modalities including meteorological forecasts, satellite imagery, numerical sensor streams, and grid interaction logs has motivated the application of multimodal learning techniques to improve time series forecasting accuracy. Unlike previous surveys that either focused narrowly on single-modality forecasting or provided only high-level overviews, this paper offers a focused and technically grounded review of multimodal learning strategies specific to renewable energy forecasting. We categorize and compare key fusion strategies (early, late, hybrid), analyze deep model architectures (e.g., co-attention and transformer-based fusion), and discuss deployment-related challenges such as data alignment, modality missingness, and interpretability. Then, this paper reviews multimodal learning techniques based on four main aspects, i.e., classify fusion techniques, examine model architectures, showcase practical application scenarios, and summarize benchmark datasets. Finally, the paper identifies existing research gaps and outlines promising future directions in this rapidly evolving field, such as self-supervised multimodal learning and adaptive fusion under uncertainty.

INDEX TERMS Multimodal learning, renewable energy forecasting, time series forecasting, data fusion, deep learning, benchmark data sets, forecast uncertainty.

I. INTRODUCTION

The global transition toward cleaner and more sustainable energy sources has accelerated the deployment of solar, wind, and hybrid energy systems [1]. Accurate forecasting of energy generation from these sources is essential to maintain grid stability, optimize operational efficiency, and enable effective participation in the energy market [2]. Unlike conventional energy generation, renewable energy sources are inherently intermittent and non-linear, driven by a complex combination of meteorological variables, geographic conditions, and temporal patterns of energy demand [3]. As a result, accurate forecasting remains a

central challenge in the reliable integration of renewable energy sources into modern power systems [4].

Traditional statistical and physics-based forecasting models often struggle to capture the non-stationary and multimodal nature of environmental data associated with solar irradiance, wind speed, and temperature fluctuations. Recent advances in data-driven methods, particularly those based on artificial intelligence (AI), have shown substantial potential in addressing these complexities. By leveraging large-scale historical datasets, AI models can effectively model non-linear dependencies and adapt to evolving temporal and spatial patterns [5], [6]. These capabilities have driven increasing adoption of machine learning and deep learning techniques for renewable energy forecasting.

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Among these advancements, multimodal learning has emerged as a powerful paradigm that integrates information from diverse sources such as satellite imagery, numerical weather prediction (NWP) models, sensor data, and historical energy production records into unified forecasting models [7]. The fusion of heterogeneous data types has shown strong effectiveness in enhancing model performance and predictive accuracy, especially when dealing with missing or noisy observations. By jointly leveraging complementary patterns across different modalities, multimodal learning enables more robust, context-aware forecasting frameworks that can generalize across geographic and temporal variability [8].

For instance, integrating spatial satellite data with temporal meteorological forecasts has been shown to significantly improve solar and wind power predictions. In a recent study, a hybrid model that combined MODIS satellite data and NWP inputs achieved a 13.2% improvement in RMSE compared to single-modality models [9]. Similarly, multimodal convolutional transformer networks have been applied to fuse geographical maps, time series weather data, and topological grid features, yielding state-of-the-art performance in wind energy forecasting tasks [10].

Despite the promising results, integrating multimodal data into forecasting systems remains challenging due to issues such as data alignment across modalities, imbalance in data granularity, and the need for scalable learning architectures that can handle high-dimensional inputs. These concerns underscore the need for structured surveys and critical assessments of recent work to help guide the development of practical, high-performance multimodal forecasting systems.

This paper addresses this gap by presenting a focused and up-to-date survey on multimodal learning approaches for renewable energy forecasting. Rather than reiterating general renewable energy principles, our emphasis is on categorizing state-of-the-art deep learning-based models, identifying core fusion strategies, and highlighting their role in improving forecasting accuracy and robustness across solar, wind, and hybrid energy systems.

To clarify the distinct contributions of this survey relative to existing literature, we introduce Table 1, which compares recent relevant surveys across key dimensions including forecasting domains covered, data modalities considered, fusion techniques discussed, deep learning models analyzed, and inclusion of benchmark datasets and evaluation metrics. This comparative summary highlights our manuscript's unique emphasis on comprehensive multimodal fusion strategies and deep learning architectures tailored specifically for renewable energy forecasting applications.

The main contributions of this survey are as follows. We provide a structured taxonomy of AI-based forecasting models with an emphasis on their multimodal data handling capabilities. We analyze the impact of various data fusion techniques early, intermediate, and late fusion, on predictive performance. We discuss the availability of benchmark datasets and the use of standardized evaluation metrics that facilitate reproducibility and fair comparison across

models. Finally, we explore remaining challenges and future directions for integrating multimodal learning in real-world forecasting applications.

The structure of this paper is organized as follows. Section II provides an essential overview of renewable energy forecasting and the motivation for multimodal learning, while avoiding redundant background. Section III details the types of data modalities relevant to renewable energy systems. Section IV outlines fusion strategies used in multimodal learning, followed by Section V, which presents representative model architectures such as CNNs, transformers, and attention-based models. Section VI presents applications in renewable energy forecasting. Section VII reviews evaluation protocols and benchmark datasets, and Section VIII discusses technical challenges. Section IX discusses future research directions. Finally, Section X concludes the paper with a summary and outlook on.

II. BACKGROUND

This section briefly introduces the renewable energy (RE) context in which multimodal learning is applied, with emphasis on the diverse data modalities (e.g., weather observations, satellite imagery, and numerical predictions) relevant to forecasting tasks. It outlines the core forecasting challenges posed by renewable systems, such as nonstationarity, missing data, and spatial heterogeneity, that motivate the use of multimodal learning. Finally, it defines multimodal learning and discusses its benefits and application challenges in the energy domain.

A. RENEWABLE ENERGY SYSTEMS AND ASSOCIATED DATA MODALITIES

Modern renewable energy systems, particularly solar PV, wind farms, and their hybrid integrations, generate substantial volumes of heterogeneous data. These include ground-based sensor measurements (e.g., global horizontal irradiance, wind speed), numerical weather prediction (NWP) outputs, satellite imagery, spatial-geographical features, and historical power generation records. Each of these modalities offers complementary perspectives for forecasting energy output. However, the inherent intermittency and variability of these systems present forecasting challenges that cannot be sufficiently addressed using any single data modality alone [16], [17], [18].

B. FORECASTING CHALLENGES AND THE NEED FOR MULTIMODAL MODELS

Time series forecasting in renewable energy systems is inherently complex due to the stochastic nature of weather-driven generation. Key challenges include: (1) high temporal and spatial variability, (2) nonlinear relationships between inputs and outputs, (3) missing or noisy data due to sensor failures, and (4) site-specific dynamics that reduce generalizability [19], [20], [21]. Traditional single-modality models, e.g., those relying solely on historical generation or on-site weather data, often fail to capture these complex

TABLE 1. Comparative summary of recent surveys on renewable energy forecasting and multimodal learning.

Survey Reference	Forecasting Domains	Data Modalities	Fusion Techniques	Deep Learning Models	Benchmarks & Metrics
Qin et al. (2022) [6]	Solar	Ground, Satellite	Hybrid	Deep Learning (DL)	RMSE
Zhang et al. (2022) [11]	Meteorological	Multi-modal Weather	Spatio-temporal Fusion	DNN	RMSE, MAE
Heo et al. (2021) [9]	Solar	Geographic, Meteorological	Early	Multi-channel CNN	RMSE
Tuncar et al. (2024) [10]	Wind	Maps, Weather	Intermediate	Transformer, CNN	Review-Based
Wang et al. (2023) [12]	Power Demand	Multimodal	Hybrid	CNN-LSTM	MAPE, RMSE
Wang et al. (2024) [13]	Solar PV	Satellite, Time Series	Multimodal	DL (unspecified)	MAE, RMSE
Jing et al. (2024) [14]	Solar	Multi-modal	Cross-modal Fusion	SolarFusionNet	RMSE
Pan et al. (2024) [15]	Solar PV	Sky Images, PV Data	Cross-modal Attention	Attention-based DL	RMSE
Wang et al. (2021) [6]	Solar, Wind	Weather, Sensor	Early, Late	LSTM, CNN	Partial
Zhou et al. (2021) [7]	Solar, Wind	NWP, Satellite	Attention-based	Transformer	Partial
This Work	Solar, Wind, Hybrid	Weather, Satellite, Sensor, Grid, Text	Early, Late, Hybrid, Attention	LSTM, CNN, Transformer, GNN, VAE	Comprehensive

interactions and may suffer from poor transferability to new regions or systems [22].

Moreover, when a forecasting model depends on a single data stream, its performance can degrade sharply during outages or anomalous conditions (e.g., sudden storms or sensor drift). Such limitations emphasize the need for robust models capable of integrating multiple data types. By doing so, models can leverage redundancy, contextual awareness, and complementary features, ultimately improving accuracy and resilience [23], [24].

C. MULTIMODAL LEARNING: DEFINITION AND APPLICATION IN ENERGY FORECASTING

Multimodal learning refers to machine learning techniques that integrate information from multiple data modalities such as time-series meteorological data, satellite imagery, and NWP outputs to produce richer and more robust representations [25]. In renewable energy forecasting, multimodal learning enables models to go beyond local sensor data and incorporate broader spatial and predictive context. For instance, cloud motion captured via satellite imagery can enhance solar forecasting when combined with irradiance measurements, while NWP-based wind maps improve predictions when fused with turbine-level telemetry [13], [26].

Fusion strategies in multimodal learning vary in complexity: early fusion concatenates raw features; intermediate fusion employs shared network layers or attention mechanisms; and late fusion aggregates model outputs. Each strategy has trade-offs in terms of interpretability, performance, and computational cost [27]. Despite challenges such as data alignment and heterogeneous resolution, multimodal models have shown promise in handling extreme conditions, adapting to new locations, and maintaining performance during partial data loss [24].

In this paper, we survey recent advances in multimodal learning for renewable energy forecasting, with an emphasis on deep fusion architectures and applications across solar, wind, and hybrid systems.

III. MODALITIES IN RENEWABLE ENERGY SYSTEMS

A. NUMERICAL SENSOR DATA

Renewable energy systems rely heavily on continuous numerical sensor measurements to monitor operational and environmental variables crucial for forecasting and control.

These sensors provide high-frequency, real-time data streams such as power output, wind speed, solar irradiance, and temperature, which form the backbone of predictive analytics in renewables.

Table 2 summarizes common sensor types, the variables they measure, and their principal applications in wind, solar, and hybrid energy systems.

Effective integration of numerical sensors requires attention to data quality (calibration, redundancy), communication protocols (Modbus, IEC 61850), and appropriate sampling rates depending on application needs.

Table 3 highlights key forecasting use cases leveraging numerical sensor data, from short-term power forecasting to fault detection and advanced feature engineering.

B. WEATHER FORECASTS

Numerical Weather Prediction (NWP) models provide structured, spatio-temporal meteorological forecasts essential for solar and wind power prediction. They solve physical equations governing atmospheric dynamics to generate forecasts of variables like temperature, humidity, wind speed, cloud cover, and solar radiation [44].

Common NWP outputs used in renewables include:

- 2 m air and surface temperature.
- Relative humidity affecting irradiance transmission.
- Wind speed/direction at turbine hub heights.
- Cloud cover by altitude layers.
- Precipitation and pressure relevant for turbine safety.
- Solar radiation fluxes such as GHI and DNI [45].

NWP forecasts are often interpolated to plant locations via geospatial methods (IDW, kriging).

Due to inherent model limitations and uncertainties, forecast errors are corrected through post-processing:

$$\text{Forecast Error} = y_{\text{true}} - \hat{y}_{\text{NWP}}$$

Machine learning methods for bias correction are increasingly used:

$$y_{\text{corrected}} = f_{\text{ML}}(\hat{y}_{\text{NWP}}, \text{auxiliary features})$$

where f_{ML} can be Random Forests, GBMs, or DNNs trained to improve accuracy [46].

NWP challenges include resolution limits, forecast drift, latency, and large data volumes [44], [52].

TABLE 2. Summary of numerical sensor types, measured variables, and applications in renewable energy systems.

Sensor Type	Measured Variables	Key Applications and Recent References
Power Output Sensors	- Real-time electrical power (kW, MW) at inverter/generator output.	- Estimating capacity factor and operational efficiency. - Real-time grid dispatch and control [28]. - Data resolution typically from 1 s to 5 min.
Wind Speed and Direction Sensors (Anemometers, Wind Vanes)	- Wind speed (m/s) at multiple heights (10 m, 60 m, 100 m). - Wind direction (degrees).	- Wind power potential estimation via cubic wind speed relation [11]. - Turbine yaw control and wake effect modeling [29]. - Inputs for wind energy forecasting and control [30], [31].
Irradiance Sensors (Pyranometers, Pyrheliometers)	- Global Horizontal Irradiance (GHI). - Direct Normal Irradiance (DNI). - Diffuse Horizontal Irradiance (DHI).	- Input to PV performance models (e.g., PVWatts, SAPM) [32]. - Solar irradiance forecasting with high sampling rates [33].
Temperature Sensors	- Ambient air temperature. - PV module surface temperature. - Equipment temperature (inverters, batteries).	- Thermal management and module efficiency correction [34], [35]. - Early fault detection by monitoring overheating [36].

TABLE 3. Key forecasting applications using numerical sensor data.

Application	Description and Techniques
Short-Term Power Forecasting	Forecast horizons from minutes to hours support grid balancing. Techniques include: • ARIMA models for stationary temporal dependencies. • LSTM neural networks for capturing long-term temporal correlations [37], [38]. • CNNs for extracting local temporal patterns in time series [39]. • Hybrid CNN-LSTM architectures combining spatial and temporal features [40], [41].
Fault Detection and Anomaly Identification	Detect anomalies by comparing observed sensor data to baseline model predictions. • Residual-based fault detection. • Autoencoder and isolation forest-based anomaly detection on multivariate streams [42].
Feature Engineering for Hybrid Models	Extract statistical (mean, variance), time-domain (lags, autocorrelation), and frequency-domain features (Fourier/Wavelet transforms) from sensor data. • Cross-sensor correlations improve joint modeling [43].

TABLE 4. Comparison of common NWP models used in renewable energy forecasting.

Model	Resolution	Update Frequency	Domain	Owner and Example Applications
ECMWF (IFS)	9 km	6-hourly	Global	European Centre for Medium-Range Weather Forecasts; widely used in Europe [47].
GFS	13 km	6-hourly	Global	NOAA (USA); basis for many global forecasts [48].
HRRR	3 km	Hourly	USA	NOAA/ESRL; turbine-level wind forecasts [49].
ICON	13 km	6-hourly	Global	Deutscher Wetterdienst (DWD); ensemble forecasting in Europe [50].
AROME	2.5 km	Hourly	Europe	Météo-France; high-resolution solar nowcasting [51].

C. SATELLITE AND IMAGE DATA

Satellite imagery provides critical spatio-temporal data for cloud cover, atmospheric dynamics, terrain shading, and surface reflectivity, supporting irradiance forecasting, terrain-aware wind modeling, and visual inspection [53].

Common satellite platforms include:

- GOES: Real-time monitoring over the Americas, 5–15 min revisit.
- Meteosat MSG: Europe, Africa, Middle East; 15 min temporal, 1–3 km spatial resolution.
- Himawari-8/9: East Asia/Oceania; 10 min revisit, multiple spectral bands.
- MODIS: High-res multispectral data for albedo and aerosol [54].
- Landsat 8/9: 30 m resolution for terrain and vegetation.

Key image processing and modeling approaches:

- **Cloud Motion Estimation:** Optical flow methods (Lucas-Kanade, Horn-Schunck) or data-driven frame prediction estimate cloud displacement:

$$I_{\text{predicted}}(t + \Delta t) = I(t) + \vec{v}_{\text{cloud}} \cdot \Delta t$$

- Digital Elevation Models (DEMs) for shading and wind modeling.
- Shadow detection and albedo correction using NIR reflectance.
- Radiative transfer models combine satellite cloud properties and solar geometry for irradiance estimation.

Satellite data applications include:

- Very short-term solar nowcasting (0–30 min) [55].
- Ramp event prediction via temporal cloud and temperature sequences.
- Terrain-aware wind micro-siting.

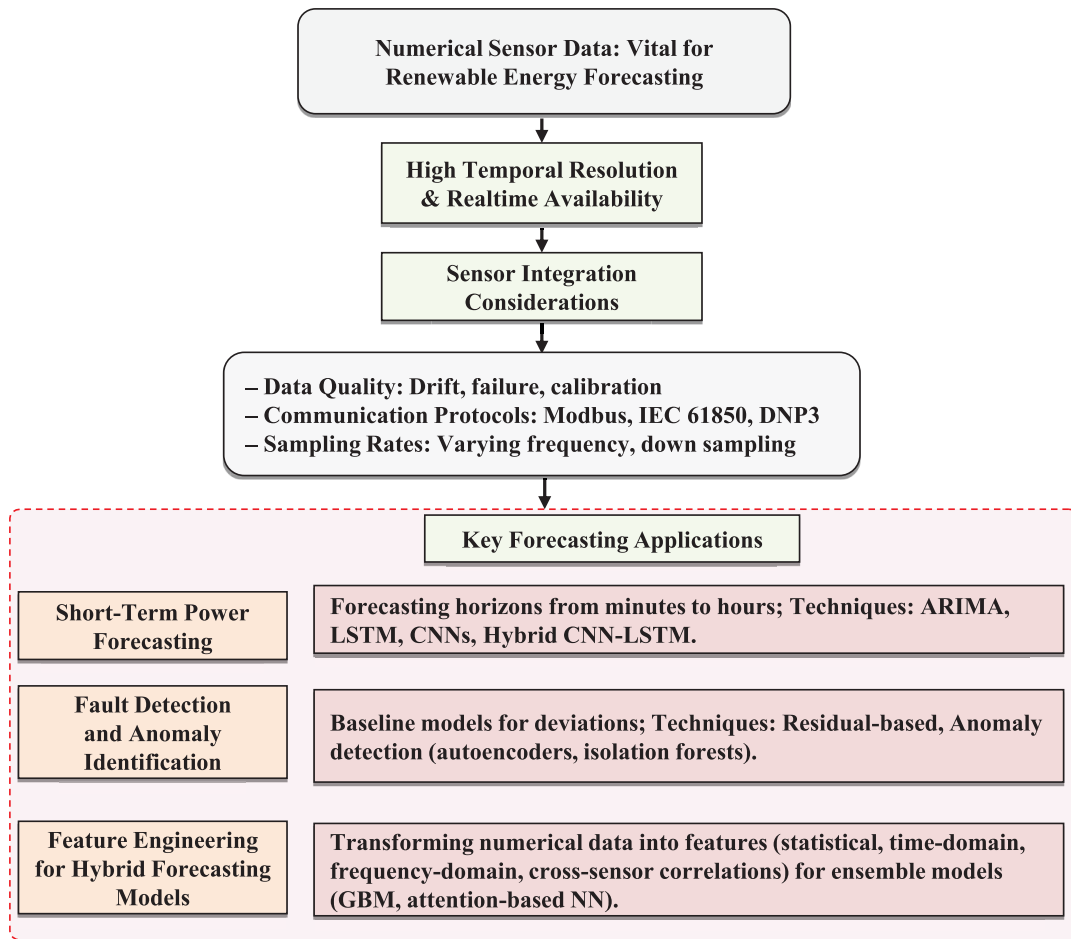


FIGURE 1. Diagram illustrating sensor integration considerations and key forecasting applications of numerical sensor data.

- Panel condition monitoring via drone or satellite imagery [56].
- Aerosol and dust storm monitoring adjusting irradiance forecasts [57].

Integration with other modalities enables:

- Fusion with NWP for cloud cover correction and hybrid modeling.
- Combining CNN-extracted image embeddings with time series features in ConvLSTM or Transformer models [58].
- Attention mechanisms dynamically weighing image vs numerical features.

Challenges include:

- Geolocation errors due to orbital drift [59].
- Cloud obscuration necessitating masking or alternate sensing [60].
- Temporal gaps from satellite revisit limitations [61].
- High data volumes requiring optimized storage and processing [62].

- Edge artifacts in optical flow reducing cloud motion reliability [63].

D. TEXTUAL DATA

Textual data provide a rich semantic layer to renewable energy forecasting by capturing contextual information from unstructured documents such as maintenance logs, incident reports, and meteorological bulletins. These data sources reveal latent patterns related to system behavior, faults, and external disturbances, complementing numerical sensor and weather data when processed using Natural Language Processing (NLP) techniques [64].

Various textual sources relevant to forecasting and fault diagnosis include: These are summarized in Table 5.

- **SCADA Maintenance Records:** Logs generated by operators or automated systems detailing routine inspections, servicing, and component replacements [65], [66].
- **Technician Notes and Alarm Logs:** On-site observations and sensor-generated warnings recorded in natural language [67].

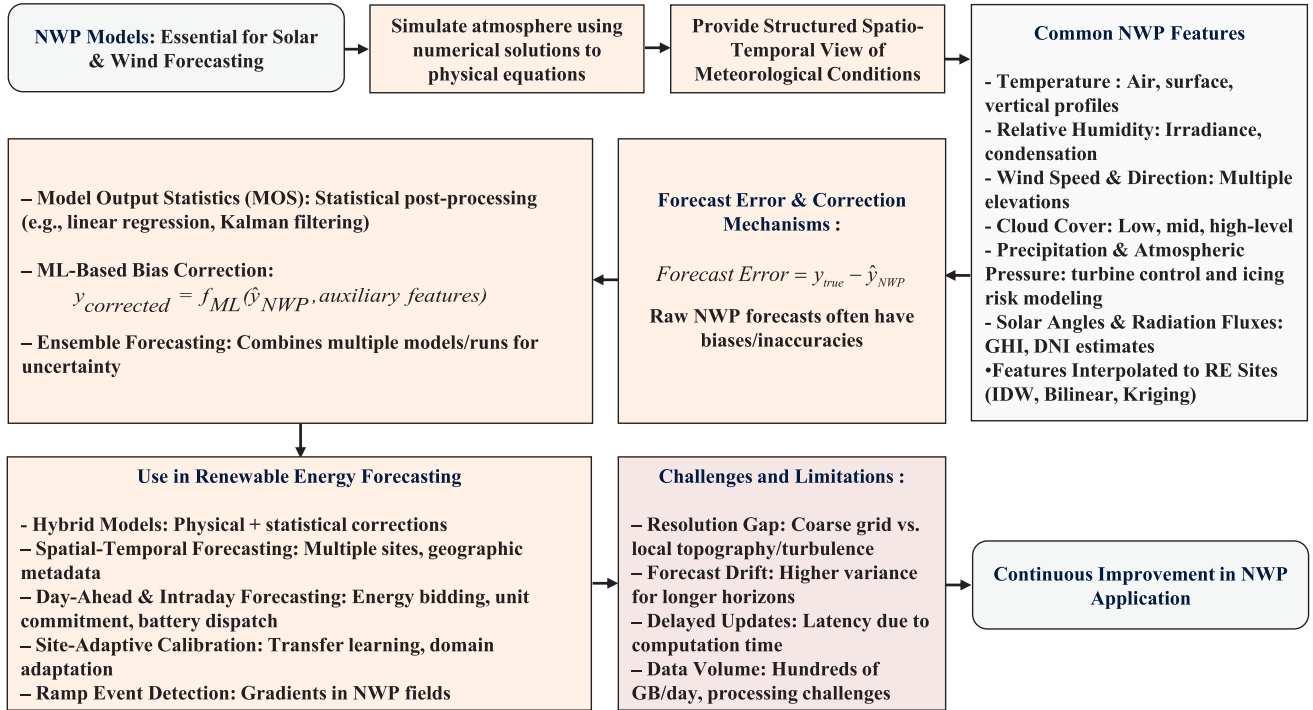


FIGURE 2. Flowchart illustrating NWP concepts and applications in renewable energy forecasting.

- **Weather Bulletins and Advisories:** Reports from meteorological agencies on extreme conditions such as storms or icing.
- **Manufacturer Reports:** Manuals, datasheets, and OEM-issued alerts relevant to system diagnostics.
- **Incident and Failure Reports:** Post-event analyses that support forensic investigations and system improvement.

These documents often exist in heterogeneous formats (e.g., PDFs, logs, HTML), requiring robust text preprocessing for effective analysis, as summarized in Table 5.

NATURAL LANGUAGE PROCESSING TECHNIQUES

NLP transforms raw text into structured information usable by forecasting models. Common techniques include:

- **Text Preprocessing:** Includes lowercasing, stop-word removal, punctuation stripping, and stemming or lemmatization.
- **Tokenization and Named Entity Recognition (NER):** Breaks down text into tokens and identifies entities such as equipment names, fault codes, or weather terms.
- **Vectorization (TF-IDF, Word2Vec):** Converts documents into numerical vectors:

$$TF\text{-}IDF(t, d) = tf(t, d) \cdot \log \frac{N}{df(t)} \quad (1)$$

where $tf(t, d)$ is the term frequency and $df(t)$ is the document frequency across N documents.

- **Topic Modeling (LDA):** Latent Dirichlet Allocation helps discover underlying themes (e.g., “equipment wear”, “storm warnings”).
- **Sentiment and Polarity Analysis:** Classifies text as positive, neutral, or negative useful in triaging fault-prone conditions.
- **Text Classification:** Uses models such as Naive Bayes, SVM, or BERT-based transformers to categorize text (e.g., fault types, risk levels).

Although textual data are not inherently time-series in nature, they play a complementary role in enhancing forecasting accuracy and interpretability [74]:

- **Early Fault Detection:** Frequent appearance of specific alarm phrases (e.g., “overheat”, “vibration alert”) may precede component failure and enhance anomaly detection.
- **Post-Hoc Model Validation:** Textual logs help interpret unexpected forecast errors by correlating with unscheduled maintenance or extreme weather events.
- **Knowledge Graph Construction:** Integrates structured and unstructured data into entity-relation graphs, enabling root-cause analysis across components, events, and conditions.
- **Multimodal Forecast Fusion:** Textual embeddings (e.g., from BERT) are fused with sensor and numerical weather prediction (NWP) data in attention-based deep learning models for hybrid inference.

TABLE 5. Summary of textual data types in renewable energy systems.

Data Type	Source	Examples	Typical Formats
SCADA Maintenance Records	Operators, automated systems	Routine inspection logs, component replacements	Text logs, CSV, database entries [68], [69]
Technician Notes and Alarm Logs	On-site technicians, sensors	Warning messages, corrective action notes	Text files, XML, PDF logs [70]
Weather Bulletins and Advisories	Meteorological agencies	Storm warnings, temperature alerts	HTML bulletins, PDFs, RSS feeds [71]
Manufacturer Reports	Equipment OEMs	Manuals, datasheets, issue alerts	PDFs, Word documents, HTML [72]
Incident and Failure Reports	Forensic teams, utilities	Post-failure analysis, blackout reports	PDFs, Word documents [73]

INTEGRATION APPROACHES

Several frameworks have been developed to incorporate textual data into forecasting pipelines:

- **Text Embeddings + Time Series Models:** Combines static textual embeddings with LSTM/GRU to model system dynamics under contextual influence.
- **Joint Multimodal Models:** Transformer-based encoders (e.g., multimodal BERT, CLIP) jointly process textual and visual inputs for decision-making.
- **Graph Neural Networks (GNNs):** Text-derived relationships are used to construct node and edge structures for GNNs modeling component interactions and failures.

CHALLENGES

Despite their promise, several challenges persist in working with textual data in renewable energy systems [64]:

- **Ambiguity and Variability in Terminology:** Different technicians or vendors may describe the same event using inconsistent vocabulary.
- **Low Label Availability:** Manual annotation of text for supervised learning is labor-intensive and often unavailable.
- **Multilingual and Code-Mixed Data:** Logs written in non-English or mixed languages (e.g., French-English in bilingual regions) complicate preprocessing.
- **Timestamp Misalignment:** Textual records may lag real-time events by hours or days, complicating synchronization with time-series data.
- **Privacy and Access Restrictions:** Maintenance records and internal reports may be confidential or subject to regulatory constraints.

IV. MULTIMODAL LEARNING STRATEGIES

A. EARLY FUSION: FEATURE CONCATENATION AT INPUT LEVEL

Early fusion is one of the simplest yet widely adopted strategies in multimodal learning. It involves combining features from multiple data modalities at the input level to form a single unified representation [27], [75], which is then passed to a downstream model. This approach assumes that all relevant information from different modalities can be

effectively captured and leveraged together without explicitly modeling modality-specific interactions.

In renewable energy forecasting, early fusion typically entails concatenating meteorological variables (e.g., wind speed, temperature, humidity), temporal descriptors (e.g., time of day, seasonal indices), and spatial features (e.g., elevation, latitude/longitude) into a single input vector. This fused vector is subsequently used to train models such as multilayer perceptrons (MLPs), random forests, or deep neural networks [76].

Despite its conceptual simplicity, early fusion faces key challenges in handling heterogeneous and unaligned data. For instance, numerical weather prediction (NWP) features are often continuous and temporally dense, whereas geographic metadata can be categorical or sparse, requiring careful preprocessing and embedding techniques to harmonize inputs [77]. Furthermore, concatenation may dilute crucial cross-modal interactions, especially in high-dimensional spaces, risking the dominance of noisy or irrelevant features.

Nevertheless, early fusion remains popular due to its computational efficiency and straightforward implementation. Recent deep learning models improve on naive concatenation by first passing each modality through dedicated encoders or embedding layers before merging, thus enhancing representational richness and facilitating learning from correlated multimodal patterns [78].

Techniques such as feature selection, dimensionality reduction (e.g., PCA, autoencoders), and regularization (dropout, batch normalization) are commonly applied to improve early fusion robustness and generalization. Figure 3 presents a redesigned flowchart of the early fusion strategy, emphasizing the separate preprocessing of modalities before concatenation, which supports clearer understanding and replicability.

B. LATE FUSION: COMBINING PREDICTIONS FROM MODALITY-SPECIFIC MODELS

Late fusion, or decision-level fusion, combines outputs from independently trained modality-specific models rather than merging raw features [75], [79]. Each modality is processed separately, and their predictions are aggregated through heuristics (e.g., averaging, weighted voting) or meta-learning methods like stacking [80].

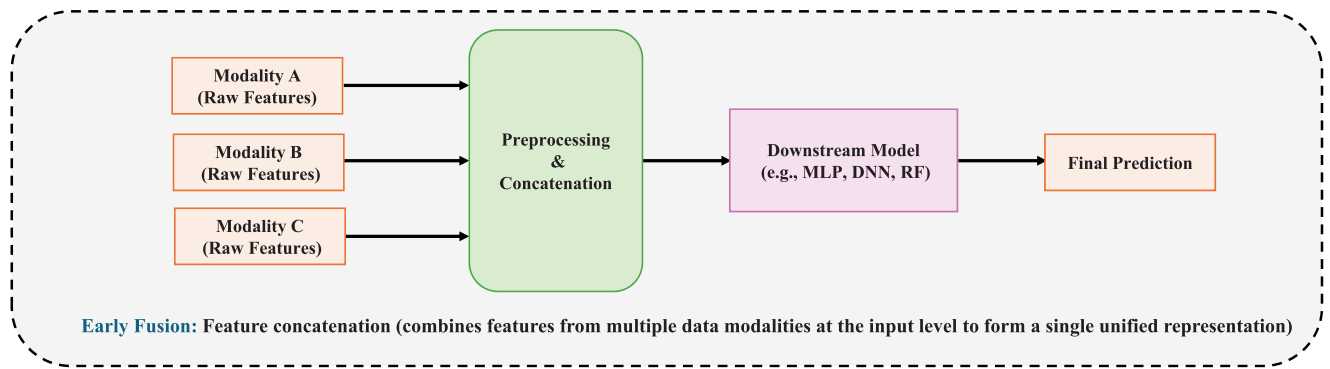


FIGURE 3. Flowchart of the early fusion learning strategy illustrating modality-specific preprocessing, feature concatenation, and unified modeling. This design supports efficient integration of diverse inputs while highlighting preprocessing steps critical for handling heterogeneous data.

In renewable energy forecasting, late fusion pipelines commonly involve separate models for meteorological variables and satellite or SCADA data, whose outputs are combined to improve final prediction reliability.

Late fusion provides significant advantages: modularity allows independent training and updating of models; specialization enables tailored architectures (e.g., CNNs for images, RNNs for sequences); and robustness arises from graceful degradation if a modality is missing or corrupted. However, late fusion requires synchronization mechanisms, such as interpolation or buffering, to reconcile different temporal resolutions across modalities [81].

While late fusion methods do not inherently learn cross-modal feature dependencies, incorporating a meta-learner to weigh predictions adaptively can partially address this limitation by exploiting context or reliability cues [80]. Careful validation and regularization are necessary to prevent overfitting in stacked or ensemble models trained on limited data.

Figure 4 shows the updated late fusion flowchart, now with explicit illustration of separate modality processing, synchronization steps, and prediction aggregation, providing clearer guidance for implementation.

C. HYBRID/INTERMEDIATE FUSION: ATTENTION-BASED AND CO-LEARNING ARCHITECTURES

Hybrid or intermediate fusion integrates modalities at latent representation levels within neural networks [75], [82]. Attention mechanisms, gating units, and shared latent spaces enable dynamic, context-sensitive interactions between modalities.

Attention-based models allow selective focus on relevant temporal or spatial features across modalities. For example, in solar forecasting, attention can emphasize satellite image regions with cloud activity while highlighting meteorological time points with rapid humidity changes.

This strategy captures conditional and complex inter-modal dependencies, adapting fusion weights dynamically based on context critical for renewable energy where relationships vary with geography and time [8]. Co-

learning approaches further enhance fusion by aligning modality-specific features via bilinear pooling or tensor fusion, promoting joint latent space discovery.

Though hybrid fusion increases model complexity and training requirements, it offers improved interpretability through attention visualization and gating activations. Figure 5 presents the enhanced hybrid fusion flowchart, clarifying intermediate feature extraction, attention modules, and joint decoding.

D. CROSS-MODAL ATTENTION AND CO-ATTENTION MECHANISMS

Cross-modal attention explicitly models interdependencies between modalities [83]. Unlike unimodal attention, it allows queries from one modality to attend to keys and values in another, enabling fine-grained contextual alignment.

In renewable energy, this could involve satellite cloud data modulating the relevance of temperature readings or topographical maps informing wind sensor data interpretation. Co-attention further symmetrizes this interaction, with modalities mutually attending to one another.

Transformer-based and multimodal BERT models have adopted these mechanisms successfully in vision-language tasks and are increasingly applied to renewable energy forecasting, showing gains in accuracy, missing data robustness, and interpretability [84].

The computational overhead of attention layers is significant, particularly for high-dimensional modalities. Sparse and hierarchical attention mechanisms help manage this complexity.

Figure 6 depicts the cross-modal attention flowchart with detailed query-key-value interactions across modalities, providing an intuitive visualization of this advanced fusion technique.

E. SELF-SUPERVISED, CONTRASTIVE, AND GENERATIVE APPROACHES (E.G., MULTIMODAL VAES)

Self-supervised learning exploits unlabeled data by predicting parts of input from others, enabling effective representation learning without extensive annotations [85]. This

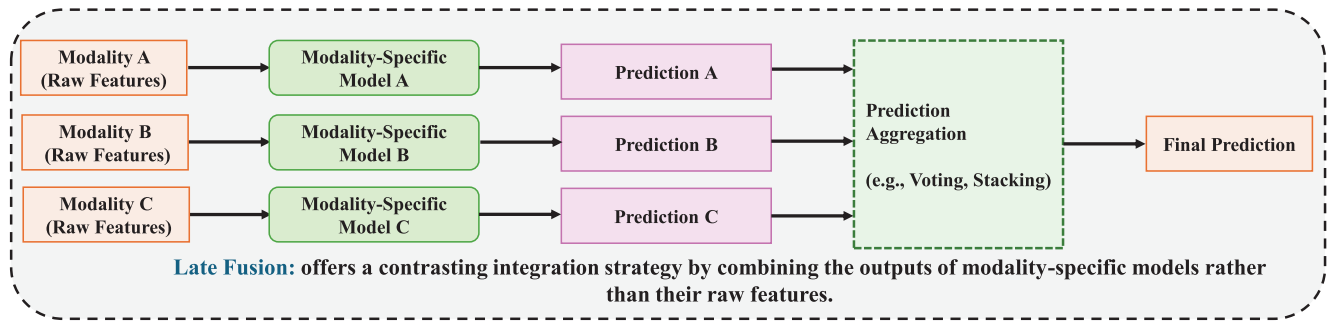


FIGURE 4. Flowchart of the late fusion learning strategy. The diagram highlights independent modality-specific modeling, synchronization of predictions, and decision-level aggregation. This modular approach enhances robustness and facilitates flexible model updates.

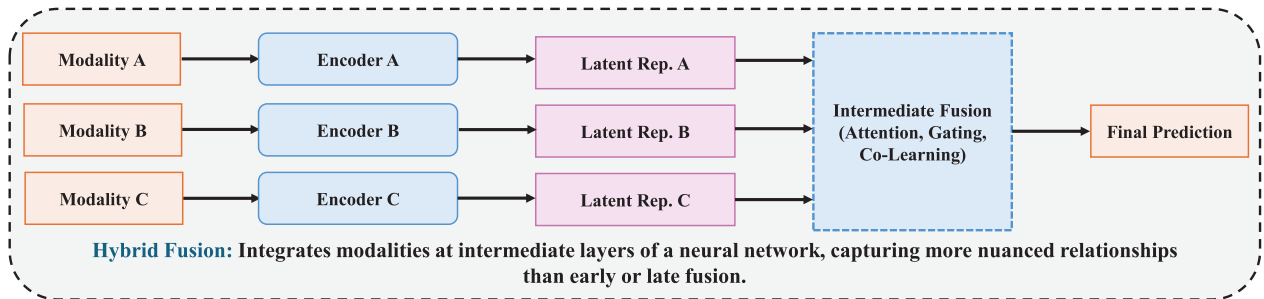


FIGURE 5. Flowchart of the hybrid/intermediate fusion learning strategy. The figure illustrates modality-specific encoders, attention-based fusion at latent layers, and joint prediction, highlighting the model's dynamic integration capacity.

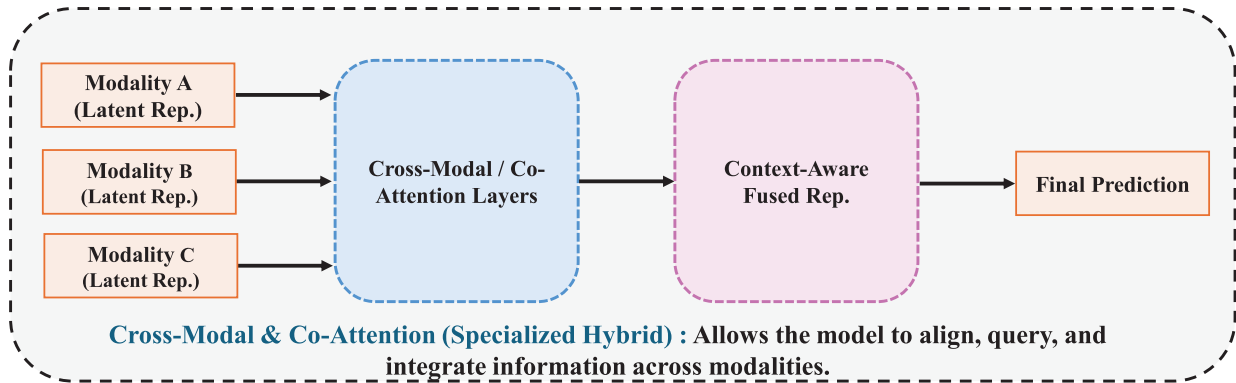


FIGURE 6. Flowchart of the cross-modal attention fusion learning strategy. The diagram shows bidirectional attention between modalities, facilitating fine-grained, context-aware feature integration.

is critical in renewable energy where labeled multimodal datasets are scarce.

Contrastive learning aligns positive modality pairs (e.g., co-occurring satellite and weather data) while separating negatives, enhancing modality-invariant feature learning. Generative models like multimodal variational autoencoders (VAEs) learn shared latent spaces for cross-modal reconstruction and data imputation, improving robustness to missing sensors [86].

These approaches support data augmentation and simulate rare events, essential for comprehensive forecasting in hybrid energy systems.

While promising, they require careful training to avoid representation collapse and demand task-specific evaluation metrics beyond conventional accuracy.

Figure 7 presents the self-supervised fusion flowchart, outlining contrastive and generative model components, emphasizing their role in learning robust multimodal embeddings.

F. PERFORMANCE COMPARISON OF FUSION STRATEGIES

Table 6 summarizes empirical forecasting performance reported in key recent studies employing early, late, and hybrid fusion methods in renewable energy applications. Metrics include Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and accuracy, allowing direct comparison across fusion approaches, datasets, and model architectures.

These results highlight that while early fusion offers simplicity, hybrid and attention-based fusion strategies often

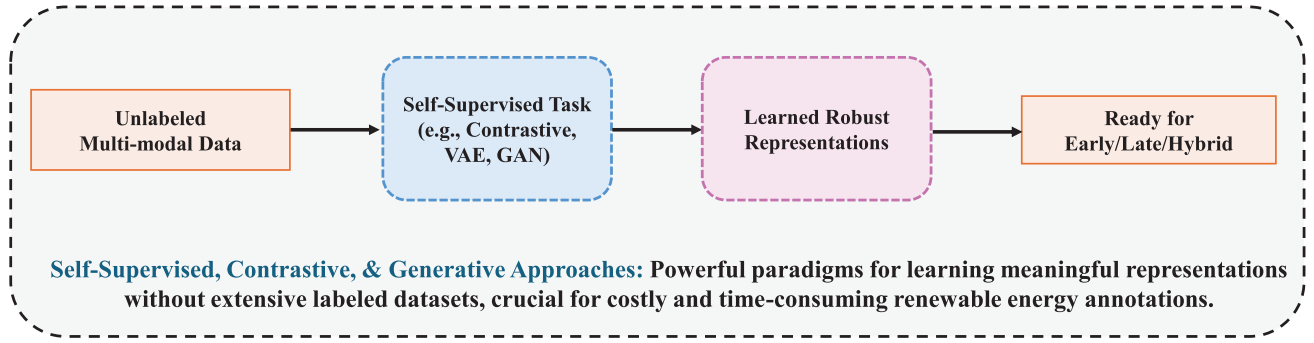


FIGURE 7. Flowchart of the self-supervised fusion learning strategy illustrating contrastive and generative model components for robust multimodal representation learning.

TABLE 6. Performance comparison of fusion types in renewable energy forecasting.

Reference	Fusion Type	Dataset	Metric	Performance
[12]	Early Fusion	Solar energy	MAP	6.12
[13]	Late Fusion	Solar energy	MSE	0.00199
[14]	Hybrid Fusion	Solar energy	nRMSE	0.280
[15]	Attention-Based	Solar energy	RMSE	2.6

achieve superior accuracy by effectively modeling complex cross-modal interactions. Late fusion, although modular and robust, may lag in predictive performance unless supplemented by adaptive weighting mechanisms. Practitioners should consider the trade-offs between complexity, interpretability, and data availability when selecting fusion techniques for renewable energy forecasting.

V. MODEL ARCHITECTURES

A. TRADITIONAL MACHINE LEARNING WITH ENGINEERED MULTIMODAL FEATURES

Traditional machine learning approaches such as linear regression, support vector machines (SVMs), and random forests have historically contributed to renewable energy forecasting by utilizing handcrafted features derived from different modalities (e.g., temperature, solar irradiance, wind speed, humidity). These features are typically concatenated into a single multimodal feature vector:

$$\mathbf{x} = [x^{(1)}, x^{(2)}, \dots, x^{(M)}]^\top,$$

where each $x^{(i)}$ is extracted from a specific modality. The model estimates a target variable y (e.g., power output) as:

$$y = f(\mathbf{x}) + \varepsilon.$$

These methods depend heavily on domain-specific feature engineering to capture relevant patterns, such as lagged variables and seasonal trends. While computationally efficient, they often fail to capture nonlinear interactions and dynamic cross-modal dependencies inherent in renewable energy systems [87]. Therefore, the focus of recent research has shifted toward deep multimodal models capable of automatic representation learning.

B. DEEP LEARNING APPROACHES FOR MULTIMODAL FORECASTING

1) CNNs FOR VISUAL METEOROLOGICAL MODALITIES

Convolutional Neural Networks (CNNs) serve as powerful encoders for processing spatial modalities such as satellite imagery, NWP grids, and sky camera feeds [62], [88]. These sources encode cloud morphology, atmospheric flow, and irradiance dispersion critical for short-term forecasting.

Given an image tensor $\mathbf{I} \in \mathbb{R}^{H \times W \times C}$, CNNs apply localized filters to learn hierarchical spatial features. The operation of a convolutional layer is defined as:

$$\mathbf{F}_{i,j,k} = \sigma \left(\sum_{m,n,c} \mathbf{I}_{i+m,j+n,c} \cdot \mathbf{K}_{m,n,c,k} + b_k \right),$$

where $\mathbf{K}_{m,n,c,k}$ denotes the k -th convolutional kernel and σ is typically ReLU. Output feature maps are passed through additional convolutional and pooling layers to extract multi-scale spatial representations.

For example, CNNs can forecast short-term solar irradiance by analyzing the movement and density of clouds in consecutive sky images [89]. In wind energy, CNNs extract spatiotemporal weather patterns from synoptic wind maps to predict downstream turbine output [90].

Advanced CNN architectures like U-Net and Inception support multi-resolution and context-aware learning. These features can be fused with non-visual time series via multimodal pipelines, enhancing model robustness and predictive accuracy [12].

2) LSTM/GRU FOR TEMPORAL MODALITIES

To capture sequential dynamics in temporal modalities (e.g., irradiance, wind speed), Long Short-Term Memory (LSTM)

and Gated Recurrent Unit (GRU) networks have been widely adopted [91], [92].

The core LSTM equations model hidden and cell states via input, forget, and output gates. These allow selective memory retention over time:

$$\begin{aligned} f_t &= \sigma(W_f x_t + U_f h_{t-1} + b_f), & i_t &= \sigma(W_i x_t + U_i h_{t-1} + b_i), \\ c_t &= f_t \odot c_{t-1} + i_t \odot \tanh(W_c x_t + U_c h_{t-1} + b_c), \\ h_t &= o_t \odot \tanh(c_t). \end{aligned}$$

GRUs simplify the structure using update and reset gates. These models are commonly used for predicting hourly/daily power outputs from multivariate historical data, often integrated in encoder-decoder setups for multi-horizon forecasts [93].

When fused with spatial modalities or symbolic inputs, LSTM/GRU modules offer temporal grounding for multimodal networks.

3) TRANSFORMER MODELS FOR CROSS-MODAL ATTENTION

Transformers utilize self-attention to model long-range temporal and cross-modal dependencies in a fully parallel fashion [94]. Unlike LSTM/GRU, they capture interactions across all time steps simultaneously.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V,$$

with query/key/value projections from the input embeddings. Temporal order is encoded via sinusoidal or learned positional encodings.

In multimodal settings, cross-attention layers enable alignment across modalities. For instance, image-derived keys and time-series-derived queries can be fused to predict energy output conditioned on spatial-temporal context [95].

Recent Transformer variants such as Informer [7] and Temporal Fusion Transformers [96] optimize for long sequences, sparsity, and interpretability. These models represent the frontier of multimodal sequence forecasting.

4) MULTIMODAL AUTOENCODERS AND VAEs FOR JOINT REPRESENTATION LEARNING

Multimodal autoencoders (AEs) and variational autoencoders (VAEs) project different input modalities into a shared latent space for representation learning and reconstruction [97]. VAEs introduce stochastic latent variables to model uncertainty and missing data.

$$\mathcal{L}_{\text{VAE}} = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})] - \text{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z})).$$

Such models are beneficial for: (i) compact cross-modal feature fusion, (ii) imputation of corrupted sensor data, and (iii) generation of missing modalities.

Hierarchical or conditional VAEs further enhance modularity and control. Attention-augmented AEs can weigh modality importance, improving robustness and interpretability [98].

5) GRAPH NEURAL NETWORKS FOR SPATIAL SENSOR TOPOLOGIES

Graph Neural Networks (GNNs) model energy systems as graphs, with nodes representing sensors or sites and edges capturing spatial or communication relations [99].

$$\mathbf{H}^{(l+1)} = \sigma\left(\tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2} \mathbf{H}^{(l)} \mathbf{W}^{(l)}\right),$$

where $\tilde{A}$ includes self-loops and $\mathbf{H}^{(l)}$ are node features. GNNs enable message passing across geographically distributed sensors to learn spatial correlations.

Graph Attention Networks (GATs) refine this by learning edge-specific attention weights. Temporal extensions integrate GNNs with RNNs or Transformers to model spatiotemporal dependencies in wind farms or PV arrays.

These networks are especially effective for large-scale, sensor-dense renewable systems where spatial topology and data quality vary dynamically [100].

VI. APPLICATIONS IN RENEWABLE ENERGY FORECASTING

A. SOLAR PV POWER FORECASTING USING IRRADIANCE, CLOUD IMAGERY, AND WEATHER DATA

Solar photovoltaic (PV) power forecasting plays a crucial role in optimizing grid operations and energy management due to the intermittent and weather-dependent nature of solar power generation. Accurate prediction relies on integrating multiple heterogeneous data sources such as irradiance measurements, satellite or ground-based cloud imagery, and meteorological variables including temperature, humidity, and wind speed.

Irradiance G directly correlates with the PV output power P . A simplified physical model for power generation can be expressed as:

$$P = \eta \cdot A \cdot G,$$

where η denotes the panel efficiency and A the panel area. However, the effective irradiance G at the PV module surface is influenced by atmospheric conditions, cloud cover, aerosol concentration, and shading effects, leading to highly variable and nonlinear behavior.

Cloud imagery provides rich spatial-temporal information on cloud movements and density. Deep learning methods, particularly convolutional neural networks (CNNs), have been employed to extract features from satellite images or ground-based sky cameras. These features capture cloud dynamics that directly affect irradiance fluctuations. Given a sequence of cloud images at times $t, t-1, \dots, t-n$, models can forecast future irradiance $G_{t+\tau}$ or directly predict PV power output at horizon τ .

Multimodal fusion approaches effectively integrate numerical weather predictions (NWP), irradiance sensors, and image-derived cloud features to enhance forecasting accuracy over various horizons from intra-hour nowcasting to day-ahead predictions [58]. These fusion frameworks often use hybrid architectures combining CNNs for image processing

and recurrent neural networks (e.g., LSTM, GRU) or Transformers for sequential weather and irradiance data.

Key challenges in this multimodal context include precise temporal and spatial data alignment, handling missing or noisy data from sensors or imagery, and ensuring model interpretability and computational efficiency for operational deployment [101].

Overall, leveraging irradiance measurements, cloud imagery, and weather data through advanced multimodal learning significantly improves the robustness and reliability of solar PV power forecasting systems, enabling better integration of solar energy into smart grids and microgrids.

B. WIND POWER FORECASTING WITH TURBINE SENSOR DATA AND NUMERICAL WEATHER PREDICTION

Wind power forecasting is essential for the reliable integration of wind energy into power grids, given the inherent variability and intermittency of wind resources. Accurate forecasts enable better scheduling, grid stability, and economic dispatch.

Forecasting models leverage a combination of turbine sensor data such as wind speed, wind direction, rotor speed, blade pitch angle, and power output and numerical weather prediction (NWP) models that provide meteorological variables at multiple atmospheric levels and spatial scales.

Short-term forecasting horizons (from minutes up to a few hours ahead) commonly employ statistical and machine learning methods trained on historical turbine measurements and near real-time sensor streams. Longer-term forecasts (hours to days ahead) predominantly rely on physical NWP models, which simulate atmospheric dynamics to generate wind field predictions.

A classical physics-inspired model describing the wind turbine power curve $P(v)$ as a function of wind speed v is given by:

$$P(v) = \begin{cases} 0, & v < v_{\text{cut-in}} \\ P_{\text{rated}} \left(\frac{v - v_{\text{cut-in}}}{v_{\text{rated}} - v_{\text{cut-in}}} \right)^3, & v_{\text{cut-in}} \leq v \leq v_{\text{rated}} \\ P_{\text{rated}}, & v_{\text{rated}} < v < v_{\text{cut-out}} \\ 0, & v \geq v_{\text{cut-out}} \end{cases}$$

where $v_{\text{cut-in}}$, v_{rated} , and $v_{\text{cut-out}}$ denote the cut-in, rated, and cut-out wind speeds, respectively, and P_{rated} is the rated power of the turbine.

Recent advances employ deep learning architectures such as Long Short-Term Memory (LSTM) networks and Transformer models to capture complex temporal dependencies and nonlinearities in turbine data and NWP forecasts [102]. These models can effectively fuse multi-source inputs, including high-resolution spatial NWP fields, turbine SCADA data, and environmental measurements, resulting in improved forecast accuracy.

Moreover, data assimilation and sensor fusion techniques integrate real-time turbine measurements with model-based forecasts to correct and refine predictions dynamically. Such

hybrid approaches enable robust and adaptive forecasting suitable for operational wind power management.

C. SHORT-TERM VS LONG-TERM GENERATION FORECASTING

Renewable energy forecasting spans a wide range of horizons [103], [104], each with distinct applications and challenges [105].

Short-term forecasting (seconds to hours ahead) is vital for real-time grid balancing, market operations, and energy trading. It demands high temporal resolution and rapid data assimilation. Models often use high-frequency sensor data and recent weather observations to predict power output with fine granularity. Techniques such as LSTM, Transformer-based models, and hybrid statistical-machine learning frameworks are commonly adopted to enhance accuracy.

Long-term forecasting (days to months ahead) supports planning, maintenance scheduling, and investment decisions. It relies heavily on coarse-resolution weather predictions and statistical models capturing seasonal and diurnal patterns [106].

Mathematically, let P_t be power at time t . Short-term forecasts minimize errors:

$$\min_{\hat{P}_{t+\tau}} \mathbb{E} \left[(P_{t+\tau} - \hat{P}_{t+\tau})^2 \right], \quad \tau \leq \text{a few hours},$$

while long-term forecasts incorporate additional uncertainty and trend components:

$$P_t = T_t + S_t + R_t,$$

where T_t is the trend, S_t seasonal component, and R_t residual noise.

Hybrid models combining physical weather simulations and data-driven methods improve accuracy across horizons, yet balancing model complexity and interpretability remains an ongoing research challenge.

D. HYBRID SYSTEMS: COMBINING SOLAR, WIND, AND BATTERY ENERGY STORAGE

Hybrid renewable energy systems integrate multiple generation sources (e.g., solar PV, wind turbines) with energy storage (e.g., batteries) to enhance reliability and smooth power output [107], [108], [109].

Forecasting in hybrid systems involves predicting each component's power output and modeling the storage system's state of charge (SoC). Let $P_{\text{solar}}(t)$, $P_{\text{wind}}(t)$, and $P_{\text{storage}}(t)$ denote solar, wind, and battery power at time t . The net output is:

$$P_{\text{net}}(t) = P_{\text{solar}}(t) + P_{\text{wind}}(t) + P_{\text{storage}}(t).$$

Battery SoC evolves as:

$$\text{SoC}_{t+1} = \text{SoC}_t + \eta_{\text{charge}} P_{\text{charge}}(t) - \frac{1}{\eta_{\text{discharge}}} P_{\text{discharge}}(t),$$

where η_{charge} , $\eta_{\text{discharge}}$ are charging/discharging efficiencies, and P_{charge} , $P_{\text{discharge}}$ are power flows.

Forecasting models for hybrid systems require multimodal data fusion to anticipate generation variability and optimize storage dispatch. Reinforcement learning and model predictive control (MPC) methods are increasingly used for optimal operation based on forecasted conditions.

Challenges include accurately modeling battery degradation, uncertainty quantification, and coordinating diverse system components in real time [110].

E. GRID-AWARE FORECASTING: RENEWABLE GENERATION AND GRID RESPONSE ESTIMATION

Grid-aware forecasting integrates renewable generation predictions with grid operational constraints and dynamics. This holistic approach enhances grid stability by anticipating not only generation fluctuations but also their impact on grid parameters such as voltage, frequency, and load flow [111].

Such models couple renewable forecasts with power flow equations and grid component models. For example, the power flow in a distribution network can be described by the AC power flow equations:

$$P_i = \sum_{j=1}^N V_i V_j (G_{ij} \cos \theta_{ij} + B_{ij} \sin \theta_{ij}),$$

$$Q_i = \sum_{j=1}^N V_i V_j (G_{ij} \sin \theta_{ij} - B_{ij} \cos \theta_{ij}),$$

where P_i , Q_i are active and reactive power injections at bus i , V_i voltage magnitudes, G_{ij} , B_{ij} conductance and susceptance, and θ_{ij} voltage angle differences. By combining forecasts of renewable output with grid state estimations, operators can preemptively manage congestion, voltage violations, and frequency excursions.

Machine learning approaches incorporating grid topology, load profiles, and renewable forecasts improve the prediction of grid responses, enabling advanced demand response and ancillary services [112].

Grid-aware forecasting represents a critical step towards resilient and efficient smart grids with high renewable penetration.

F. REAL-WORLD DEPLOYMENTS OF MULTIMODAL MODELS IN RENEWABLE ENERGY SYSTEMS

The practical application of multimodal learning techniques in operational renewable energy systems is gaining momentum worldwide. Several pilot projects and commercial deployments have demonstrated the value of integrating diverse data sources such as sensor measurements, satellite imagery, textual data, smart meter readings, and weather forecasts to enhance forecasting accuracy and support grid management decisions. Below, we highlight representative case studies from key regions:

- **Japan (Kyushu):** A multimodal ensemble deep learning framework combining CNN–LSTM was deployed to forecast day-ahead electricity prices and generation using a fusion of weather data, calendar data, and power system variables. This approach yielded enhanced prediction performance ($R^2=0.787$, $MAE=1.936$, $RMSE=2.630$), and demonstrated robustness in handling zero-price events [113].

Western Australia: A deep learning-based short-term forecasting system used LSTM models trained on smart meter data energy import/export, rooftop PV generation, consumption and outdoor temperature. The model showed superior performance compared to classical baselines in forecasting renewable energy flows on granular time scales (seconds to minutes) [114].

China: Multimodal fusion of time series and text data (including policy documents and reports) has been utilized in CNN-LSTM architectures to improve medium-range forecasts. This hybrid approach significantly enhanced accuracy and supported power system planning and the national low-carbon transition [12].

- **South Korea:** For long-term projections up to 2035, a bottom-up forecasting method based on multimodal inputs (government plans, project data, historical production records) was employed. The combined use of logistic growth, diffusion models, and regression provided detailed estimates of renewable energy development by source, guiding national policy targets [115].

Table 7 summarizes these deployments, outlining the data modalities used, the forecast horizon addressed, the adopted techniques, and the benefits achieved.

These real-world examples highlight how multimodal learning is moving beyond research laboratories into operational environments, proving its practical value in addressing the challenges of renewable energy variability and grid integration. However, challenges remain in data quality, model interpretability, and deployment scalability, which continue to motivate ongoing research.

VII. EVALUATION METRICS AND BENCHMARK DATASETS

A. FORECASTING METRICS: RMSE, MAE, MAPE, NRMSE

Accurate evaluation of renewable energy forecasting models requires appropriate error metrics that reflect prediction quality and robustness. Common metrics include Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and Normalized RMSE (NRMSE) [116]. The RMSE is defined as:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2},$$

where y_i is the true value, $\hat{y}_i$ is the predicted value, and N is the number of samples. RMSE penalizes larger errors more severely, making it sensitive to outliers. MAE measures

TABLE 7. Some real-world deployments of multimodal learning in renewable energy systems.

Region	Data Modalities	Forecast Horizon	Model/Technique	Reported Benefits
Japan (Kyushu)	Day-ahead prices, system prices, power generation, weather, calendar	Day-ahead	CNN-LSTM with ensemble learning [113]	$R^2=0.787$, MAE=1.936, RMSE=2.630; handles zero-price events effectively
Western Australia	Smart meter data (energy import/export, rooftop PV generation, consumption), outdoor temperature	Seconds to minutes	Optimized LSTM model [114]	Enhanced accuracy over classical and ML models in high-frequency forecasts
China	Time series (SCADA), policy reports, weather	Short to medium term (up to 2 years)	CNN-LSTM with multimodal fusion [12]	Improved forecasting accuracy and planning guidance; effective data fusion
South Korea	Planning data, renewable project records, historical production	Long-term (up to 2035)	Logistic growth, diffusion, regression [115]	Projections by source; supports renewable policy targets

average absolute errors:

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|,$$

providing a linear measure of error magnitude less sensitive to large deviations than RMSE. MAPE expresses errors as percentages:

$$MAPE = \frac{100\%}{N} \sum_{i=1}^N \left| \frac{y_i - \hat{y}_i}{y_i} \right|,$$

which is intuitive but can be unstable when true values y_i are close to zero.

NRMSE normalizes RMSE by the range or mean of true values:

$$NRMSE = \frac{RMSE}{y_{\max} - y_{\min}} \quad \text{or} \quad \frac{RMSE}{\bar{y}},$$

allowing comparison across different datasets or scales. The first form is typically used when the range ($y_{\max} - y_{\min}$) is relevant to daily cycles or capacity limits, while the second form is suited to relative performance comparisons over time or across systems.

B. DATASET QUALITY AND MODALITY COMPLETENESS

The performance and generalizability of forecasting models strongly depend on the quality of training and evaluation datasets. Several factors must be considered to ensure reliable model development and deployment, including:

- **Data Accuracy and Precision:** Sensor calibration and measurement errors can introduce noise, biasing forecasts. Ensuring high-quality instrumentation and data cleaning is vital.
- **Sampling Rate and Temporal Resolution:** High-frequency data captures rapid fluctuations but increases computational burden. Forecasting horizons dictate suitable temporal resolutions.
- **Completeness and Missing Data:** Multimodal datasets often suffer from missing values due to sensor faults or transmission errors. Strategies like imputation or robust modeling are needed.

- **Synchronization Across Modalities:** Temporal alignment of heterogeneous data sources (e.g., weather maps, sensor readings, images) is necessary to preserve correlations and improve fusion.
- **Dataset Size and Diversity:** Sufficient volume and variety of scenarios (seasonal changes, geographic diversity) improve model robustness.

To ensure reproducibility and scalability, benchmarks should report metadata completeness, alignment statistics, and noise levels across modalities. Tools such as modality correlation matrices and temporal heatmaps are increasingly used for dataset validation.

C. COMMON BENCHMARKS AND COMPARATIVE SUMMARY

1) SOLAR ENERGY DATASETS

Solar and wind energy forecasting benefit greatly from datasets that include historical records of energy output alongside relevant meteorological variables such as temperature, cloud cover, humidity, wind speed, and wind direction. These datasets support both deterministic and probabilistic forecasting models under various climatic and temporal conditions. Additionally, hybrid renewable energy systems require multimodal datasets capturing interactions among diverse energy sources to enable advanced forecasting using data fusion and multimodal learning approaches.

Table 8 presents dataset properties such as temporal resolution, size, and geographic coverage, as well as the specific modalities contained within each dataset. This comprehensive view aids researchers in selecting appropriate data for their forecasting tasks and understanding the scope of multimodal information available. Notably, datasets like NREL and SolarAnywhere provide diverse modalities including satellite imagery and ground sensors, enabling multimodal fusion approaches. Hybrid datasets such as Pecan Street and RENES capture complex interactions between multiple renewable sources and storage, facilitating advanced research on integrated systems. This enriched benchmark section supports clearer comparative analysis and guides future developments in renewable energy forecasting.

TABLE 8. Benchmark datasets for renewable energy forecasting: Solar, wind, and hybrid systems with properties, modalities, and key usage.

Solar Energy Datasets				
Dataset	Description	Properties	Modalities Included	Key Usage / References
GEFCom2014	Dataset from Global Energy Forecasting Competition 2014 featuring high-resolution solar power and weather data.	Temporal resolution: hourly; Size: regional US sites; Geographic coverage: multiple zones	Solar power output, meteorological variables (irradiance, temperature, humidity), numerical weather predictions (NWP)	Widely used for probabilistic solar forecasting benchmark [117]
NREL Datasets	Provided by U.S. National Renewable Energy Laboratory; includes ground-based and satellite solar irradiance and PV output data.	Temporal resolution: 1-min to hourly; Size: multiple sites; Geographic: US-wide	Ground sensor irradiance, satellite imagery, PV power output, weather forecasts	Applied in multi-modal forecasting and model evaluation [118]
SolarAnywhere	Commercial dataset offering satellite-derived solar irradiance with global coverage.	Temporal resolution: 5-min to hourly; Geographic: global; Large-scale	Satellite imagery, irradiance estimates, historical weather data	Used in hybrid forecasting and satellite-data-based modeling [119]
Solar-Energy Dataset (UCI)	Public ML benchmark dataset with long-term solar irradiance and power output records.	Temporal resolution: hourly; Single location; Moderate size	Solar irradiance, power output, temperature	Commonly used for ML benchmarking and baseline model development [120]
Wind Energy Datasets				
Wind Integration National Dataset (WIND)	NREL dataset containing wind turbine time-series and meteorological data.	Temporal resolution: 5-min; Multi-site; US-wide coverage	Wind speed/direction, turbine power, atmospheric data	Standard benchmark for spatiotemporal wind forecasting [121]
ECL Datasets	Data from utilities and research institutions combining turbine sensors and weather data.	Temporal resolution: varies; Multi-turbine; Regional	Turbine SCADA, weather station data	Used for ML and hybrid wind power forecasting [122]
GEFCom2012	Dataset for Global Energy Forecasting Competition 2012 focusing on wind data.	Temporal resolution: hourly; Multiple sites	Wind power output, meteorological variables	Benchmark for probabilistic wind forecasting [123]
Hybrid Renewable Energy Datasets				
Pecan Street Dataset	Residential/commercial consumption with solar generation and battery storage data.	Temporal resolution: 1-min to hourly; Thousands of buildings; US-based	Electricity consumption, solar PV generation, storage levels	Applied in hybrid forecasting, demand response, energy disaggregation [124]
RENES Dataset	Controlled testbed integrating solar, wind, and storage with multimodal sensor measurements.	Temporal resolution: high-frequency; Small-scale experimental setup	Solar, wind generation, storage status, weather sensors	Enables multimodal fusion model evaluation in hybrid systems [125]

VIII. CHALLENGES IN MULTIMODAL FORECASTING FOR RENEWABLE ENERGY

A. SPATIOTEMPORAL RESOLUTION MISMATCH ACROSS MODALITIES

One of the principal challenges in multimodal renewable energy forecasting arises from the mismatch in spatial and temporal resolution among the diverse data modalities. Renewable energy systems integrate heterogeneous data sources, including satellite imagery, weather station measurements, sensor arrays on turbines and solar panels, and numerical weather prediction outputs. Each of these data streams may differ substantially in spatial granularity and temporal frequency [126].

For example, satellite imagery might be available only every 15 minutes to an hour and cover broad geographic regions, whereas turbine sensor data can be collected at sub-second intervals but localized to individual units. Similarly, weather station data might provide point measurements at hourly or sub-hourly intervals, while NWP models output

gridded forecasts at resolutions ranging from a few kilometers to tens of kilometers, updated every few hours.

This disparity leads to difficulties in fusing data effectively. Direct concatenation or naive interpolation can introduce artifacts or lose critical information. Temporal mismatch may cause delays or phase shifts in the signal alignment, reducing forecasting accuracy. Spatial resolution differences complicate modeling, as fine-grained local effects might be smoothed out in coarse-grained data, and vice versa.

To address these issues, various strategies have been proposed. Spatial and temporal interpolation methods, such as kriging or spline-based approaches, can resample data to common grids or timestamps. More advanced methods exploit spatiotemporal neural networks that learn to represent different resolutions jointly. Furthermore, hierarchical or multi-scale models process modalities at their native resolutions and integrate information through learned attention mechanisms. Nonetheless, spatiotemporal resolution mismatch remains a significant bottleneck in building

robust multimodal forecasting models, requiring careful pre-processing, domain knowledge, and adaptive modeling techniques.

B. DATA ALIGNMENT, MISSING MODALITIES, AND ASYNCHRONOUS UPDATES

Multimodal forecasting systems often face challenges related to data alignment and completeness. Since different sensors and data providers operate independently, datasets may not be synchronized perfectly in time or may contain gaps due to sensor failures, communication errors, or maintenance downtime.

Asynchronous updates mean that some modalities update more frequently or irregularly compared to others. For instance, weather satellite passes may be periodic but sparse, whereas ground-based sensors can have continuous streams. Combining these requires temporal alignment strategies such as resampling, buffering, or window-based aggregation.

Missing modalities pose another challenge: in practical deployments, one or more data streams may become temporarily unavailable. Models trained on complete multimodal inputs may perform poorly or fail when some inputs are missing, limiting real-world applicability.

Handling these issues requires robust data imputation and missing data modeling techniques. Statistical imputation methods, such as mean or interpolation-based filling, are simple but may not capture complex dependencies. More advanced approaches employ machine learning models trained to reconstruct missing modalities from available data, including generative models like variational autoencoders or adversarial networks.

Moreover, asynchronous data streams call for flexible model architectures capable of processing variable-length sequences and conditional inputs. Recurrent networks with masking, attention mechanisms that weigh inputs dynamically, and multi-task learning frameworks can improve robustness [127]. Effective data alignment and missing modality handling are critical to maintain forecasting accuracy and reliability, especially in operational energy systems where sensor faults and network issues are common.

C. COMPUTATIONAL COMPLEXITY OF DEEP MULTIMODAL MODELS

The integration of multiple high-dimensional modalities in renewable energy forecasting often leads to complex deep learning models with large numbers of parameters and heavy computational demands. Models such as convolutional neural networks (CNNs) for image processing, recurrent neural networks (RNNs) or transformers for temporal data, and graph neural networks (GNNs) for spatial relationships must be combined effectively, increasing architectural complexity [128], [129].

Training these models requires extensive computational resources, including high-memory GPUs or clusters, pro-

longed training times, and careful hyperparameter tuning. This complexity can be prohibitive for researchers and practitioners with limited access to such infrastructure.

At inference time, computational cost also affects real-time deployment feasibility. Forecasting systems used by grid operators or energy traders must produce timely predictions; excessively heavy models may introduce latency or require expensive hardware upgrades [126].

Strategies to reduce computational complexity include model compression techniques such as pruning, quantization, and knowledge distillation, which simplify trained models without significant accuracy loss. Architectural innovations like lightweight convolutional blocks, efficient attention mechanisms, and parameter sharing also help [129]. Additionally, modular architectures where modality-specific encoders run in parallel, with a lightweight fusion module, balance complexity and performance. Edge computing and distributed processing approaches can further alleviate centralized computational burdens [130]. Despite these advances, computational complexity remains a core challenge in scaling multimodal forecasting systems to large geographic regions and fine temporal resolutions.

D. LACK OF STANDARDIZED MULTIMODAL BENCHMARKS

Another major obstacle in advancing multimodal renewable energy forecasting is the absence of widely accepted standardized benchmark datasets and evaluation protocols that encompass multiple data modalities. While numerous single-modality datasets exist for solar or wind forecasting individually, fewer datasets integrate multiple modalities with consistent annotation and accessibility [131].

This lack of benchmarks hinders the fair comparison of competing algorithms and slows progress. Researchers often rely on proprietary or limited datasets, leading to fragmented efforts and difficulties reproducing results.

Creating comprehensive multimodal benchmark datasets requires coordinated efforts to collect, preprocess, and release synchronized multimodal data including satellite imagery, ground sensors, meteorological variables, and historical power outputs. These datasets should cover diverse geographic regions, temporal scales, and weather regimes to enable robust generalization. Furthermore, standardized evaluation protocols specifying forecasting horizons, train-test splits, and metrics are needed to ensure comparability. Public competitions and challenges focused on multimodal forecasting could catalyze community engagement [126].

Efforts in other domains, such as multimodal video processing and medical data analysis, provide useful paradigms. However, the domain-specific constraints and complexities of renewable energy systems require a customized benchmark design. Addressing the lack of standardized multimodal benchmarks is crucial to foster transparent, reproducible research, and accelerate innovation in this field.

E. INTERPRETABILITY AND TRUST IN AI-BASED ENERGY FORECASTING

The deployment of AI models for renewable energy forecasting in operational settings requires not only accuracy but also interpretability and trustworthiness. Complex deep multimodal models are often regarded as “black boxes,” limiting user understanding of their decision processes and hampering acceptance by grid operators and energy stakeholders [132].

Interpretability is essential for model debugging, regulatory compliance, and ensuring model robustness under unexpected operating conditions. Energy system operators need to trust that forecasts are reliable and understand the influence of various inputs.

Several approaches have been developed to improve interpretability of multimodal models. Attention mechanisms can highlight which modalities or features are most influential for a given prediction. Post-hoc explanation methods, such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations), provide local feature importance.

Model simplification through surrogate models can aid human comprehension, though sometimes at the expense of accuracy. Incorporating domain knowledge and physically interpretable modules enhances trustworthiness in energy forecasting environments [133].

Furthermore, uncertainty quantification, such as prediction intervals or Bayesian approaches, communicates confidence levels, helping users make informed decisions [134].

Building interpretability and trust into AI-based renewable energy forecasting models remains an ongoing challenge but is critical for practical deployment and stakeholder acceptance.

F. COMPARATIVE ANALYSIS OF DATASETS AND EVALUATION METRICS

Table 9 summarizes prominent datasets such as NREL, GEFCom, and SolarDB, detailing the included modalities, forecasting horizons, evaluation metrics employed, and reported performance outcomes from key studies in the literature. This synthesis facilitates understanding of dataset characteristics, typical use cases, and benchmarking results, thereby aiding fair comparison and reproducibility across studies. Notably, datasets vary widely in geographic coverage, temporal resolution, and modality fusion, impacting model generalizability. Standardized evaluation protocols remain scarce, and reported metrics include RMSE, MAE, MAPE, and skill scores depending on the forecasting task. This table highlights trends and gaps, underscoring the need for community-driven benchmark development and shared evaluation frameworks.

This comparative analysis highlights the current fragmented landscape and motivates the development of standardized multimodal datasets and evaluation protocols to accelerate research and ensure fair model comparison.

IX. FUTURE RESEARCH DIRECTIONS

A. FOUNDATION MODELS TRAINED ON LARGE-SCALE MULTIMODAL ENERGY DATA

The rapid advancement of foundation models large-scale pretrained neural networks capable of handling diverse tasks with minimal fine-tuning presents a promising frontier for renewable energy forecasting. These models, exemplified by architectures like GPT, BERT, or vision-language transformers, demonstrate remarkable ability to learn rich, transferable representations from massive datasets spanning multiple modalities [137].

Applying foundation models to renewable energy systems entails pretraining on vast and heterogeneous multimodal datasets that include satellite imagery, weather observations, sensor telemetry, historical power outputs, and grid operational data. Such models can capture complex cross-modal interactions, spatiotemporal dependencies, and underlying physical phenomena in a unified latent space [8].

The benefits include improved generalization across geographic regions and weather patterns, reduction in data annotation effort through self-supervised learning objectives, and enhanced adaptability to downstream tasks such as anomaly detection, fault diagnosis, and scenario simulation.

Challenges remain in curating sufficiently large and diverse datasets, designing architectures that effectively fuse multimodal inputs, and managing the computational cost of training and deploying foundation models. Moreover, domain adaptation techniques must be developed to tailor foundation models pretrained on general data to specific renewable energy applications and regions.

Research on foundation models in this domain could dramatically transform forecasting accuracy and reliability, enabling more resilient and efficient energy systems.

B. FEDERATED MULTIMODAL LEARNING ACROSS RENEWABLE PLANT OPERATORS

Renewable energy data is often siloed within individual plant operators due to privacy, commercial sensitivity, or regulatory restrictions, limiting the availability of large-scale datasets for centralized training. Federated learning offers a paradigm to collaboratively train multimodal forecasting models across multiple distributed data owners without sharing raw data [138].

In federated multimodal learning, each operator trains a local model on its own multimodal dataset (e.g., sensor readings, local weather data, satellite imagery). Periodically, model updates or gradients are aggregated centrally or in a decentralized manner to update a global model. This approach preserves data privacy while leveraging the collective knowledge from diverse geographic regions and system configurations.

Technical challenges include handling heterogeneous data distributions across operators, synchronizing updates when modalities differ or are incomplete, and designing communication-efficient protocols to reduce overhead [139].

TABLE 9. Summary of common renewable energy forecasting datasets, modalities, metrics, and reported results.

Dataset	Modalities Included	Forecasting Horizon	Evaluation Metrics	Reported Results / References
NREL Solar Data	Solar irradiance, power output, weather station data	Hourly, daily	RMSE, MAE, MAPE	[118]
GEFCom2012	Wind speed, power generation, meteorological forecasts	Hourly, sub-hourly	RMSE, skill score	[135]
SolarDB	Satellite imagery, weather variables, power data	15 min, hourly	MAE, RMSE	[136]
RENES Dataset	Solar, wind generation, storage, weather sensors	Seconds to minutes	RMSE, MAE, Pearson R	[125]
SolarAnywhere	Satellite-derived irradiance, weather reanalysis	5-min, hourly	MAE, nRMSE, Bias	[119]

Further, multimodal federated learning requires advanced aggregation methods that can reconcile discrepancies in modalities and sensor types, and robust techniques to handle missing or corrupted local data.

Successful development of federated multimodal frameworks would enable cross-plant knowledge transfer, improve forecasting accuracy, and promote collaborative innovation without compromising data confidentiality.

C. FEW-SHOT AND TRANSFER LEARNING IN MODALITY-SCARCE ENVIRONMENTS

While many renewable energy plants produce rich multimodal data, others may have limited sensing infrastructure or face operational constraints that restrict the availability of certain modalities. Few-shot and transfer learning methodologies aim to overcome this scarcity by leveraging knowledge learned from data-rich domains to improve performance in data-poor settings. Few-shot learning techniques train models to generalize from a very small number of labeled examples, often through meta-learning or metric learning approaches. For instance, a model pretrained on multimodal data from large solar farms could be adapted rapidly to a smaller or newly commissioned site with minimal additional data.

Transfer learning enables the reuse of pretrained feature extractors or entire models on related tasks or domains, reducing training time and data requirements. In the context of modality scarcity, transfer learning can help models compensate for missing inputs by relying on correlated or substitute modalities [140].

Developing robust few-shot and transfer learning techniques tailored to multimodal renewable energy forecasting requires addressing domain shift issues, modality mismatch, and temporal nonstationarities. Furthermore, interpretability and uncertainty estimation are crucial to ensure reliable adaptation.

Advances in this area would broaden the applicability of sophisticated forecasting models to diverse operational scenarios, accelerating the deployment of AI in renewable energy systems globally.

D. EXPLAINABLE AND UNCERTAINTY-AWARE MULTIMODAL MODELS

For AI-driven renewable energy forecasting systems to gain widespread acceptance, it is essential to develop models

that not only provide accurate predictions but also offer explanations and quantify uncertainty in their outputs.

Explainable AI (XAI) methods elucidate how multimodal inputs contribute to forecasts, enhancing user trust and facilitating model debugging. Techniques such as attention visualization, feature importance scoring, and concept-based explanations enable operators to understand the decision-making process of complex deep learning models [141].

Simultaneously, uncertainty-aware modeling quantifies confidence levels associated with predictions, distinguishing between aleatoric uncertainty (inherent data noise) and epistemic uncertainty (model knowledge gaps). Bayesian neural networks, ensembles, and probabilistic deep learning approaches provide principled frameworks for uncertainty estimation.

Incorporating explainability and uncertainty quantification is especially critical in safety-critical or economically sensitive renewable energy operations where decisions based on forecasts impact grid stability and financial planning.

Future research should focus on integrating these aspects seamlessly into multimodal architectures, balancing interpretability with predictive performance, and developing visualization tools tailored to energy system operators.

E. INTEGRATION OF PHYSICS-INFORMED LEARNING WITH DATA-DRIVEN FUSION

Renewable energy forecasting benefits from combining physics-based models grounded in domain knowledge with data-driven AI techniques that learn patterns from data. Physics-informed machine learning (PIML) incorporates physical constraints, conservation laws, or differential equations into the training of neural networks, improving model generalization and interpretability.

Integrating PIML with multimodal fusion involves designing architectures that incorporate prior knowledge about atmospheric dynamics, solar radiation physics, or turbine aerodynamics alongside multimodal observational data.

For instance, differential equation solvers or simplified energy balance models can be embedded as layers within deep networks or used as regularization terms in loss functions. This fusion helps mitigate overfitting, enforce physically plausible outputs, and enable extrapolation to unseen conditions [142].

Key challenges include representing complex physical phenomena at scales compatible with neural network train-

ing, handling noisy or incomplete data, and balancing the influence of physics-based constraints versus learned features.

Advancing the synergy between physics-informed learning and multimodal data fusion promises more accurate, robust, and trustworthy renewable energy forecasts, bridging the gap between traditional numerical models and modern AI techniques [143].

X. CONCLUSION

Accurate forecasting of renewable energy generation remains a cornerstone for the efficient management and operation of modern power systems. This survey has comprehensively examined the emerging paradigm of multimodal forecasting frameworks tailored for renewable energy systems such as solar photovoltaic, wind farms, and hybrid installations. By leveraging heterogeneous data sources, including meteorological measurements, satellite imagery, sensor telemetry, and grid operational data, multimodal approaches provide a more holistic understanding of the underlying dynamics that govern renewable power output.

Traditional single-modality forecasting models often struggle to capture the complex spatiotemporal patterns and nonlinear relationships present in renewable energy systems. Multimodal learning frameworks address these challenges by integrating complementary information across modalities, resulting in enhanced predictive accuracy and robustness. For instance, the fusion of irradiance measurements with cloud imagery and weather data significantly improves solar PV power forecasting, while the incorporation of turbine sensor data and numerical weather predictions advances wind power estimates. The flexibility of multimodal architectures, including early fusion, late fusion, and hybrid strategies, enables tailored solutions that accommodate diverse data availability scenarios and application needs.

The key advantages of integrating multiple modalities extend beyond accuracy improvements. Multimodal frameworks inherently enhance resilience to missing or noisy data, as models can rely on redundant information channels. They also enable the exploitation of cross-modal correlations and the discovery of latent factors driving renewable energy variability. Recent developments in deep learning architectures such as convolutional neural networks, recurrent neural networks (RNNs), transformers, variational autoencoders, and graph neural networks have demonstrated impressive capabilities in capturing intricate multimodal dependencies. Moreover, advances in foundation models, federated learning, and physics-informed AI point to a promising future where scalable, collaborative, and interpretable forecasting systems will become the norm.

Despite these promising developments, several critical challenges remain. Issues such as spatiotemporal resolution mismatch, asynchronous and incomplete data streams, high computational costs, and the lack of standardized multimodal benchmarks require ongoing research and innovation.

Furthermore, to fully realize the potential of multimodal forecasting in real-world renewable energy systems, it is imperative to develop robust models that balance predictive performance with interpretability and uncertainty quantification. Explainable AI techniques and uncertainty-aware frameworks will be essential to build operator trust and enable informed decision-making in grid management and energy markets.

To guide future research, we explicitly highlight the following key open challenges and promising directions:

- **Standardized Benchmark Datasets and Evaluation Protocols:** There is a critical need to develop and adopt comprehensive multimodal datasets with consistent annotations and standardized evaluation metrics to enable fair comparison, reproducibility, and accelerated progress in the field.
- **Spatiotemporal Resolution Mismatch and Asynchronous Fusion:** Future work must focus on designing advanced data fusion methods that can effectively reconcile heterogeneous spatial and temporal granularities, as well as handle asynchronous and incomplete data streams, to improve robustness and accuracy.
- **Interpretability and Uncertainty Quantification:** Developing explainable multimodal forecasting models that provide transparent reasoning and well-calibrated uncertainty estimates is essential for practical deployment and gaining trust among grid operators and energy stakeholders.
- **Federated and Privacy-Preserving Learning:** As data privacy concerns grow, expanding federated learning frameworks to multimodal renewable energy datasets will enable collaborative model training across organizations without sharing raw sensitive data.
- **Integration of Physics-Informed and Data-Driven Approaches:** Hybrid modeling paradigms that embed physical knowledge of renewable systems within data-driven multimodal architectures offer a promising route to improve generalization, interpretability, and scientific consistency.

These directions collectively form a roadmap to tackle current limitations and harness the full potential of multimodal learning for sustainable, reliable, and efficient renewable energy forecasting.

In summary, the fusion of diverse data modalities through advanced machine learning techniques represents a transformative approach to renewable energy forecasting. Continued research addressing the outlined gaps will be crucial to support the accelerating integration of renewables, optimize grid operations, and foster the transition to a resilient and low-carbon energy future.

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